Emergency Locator Transmitter (ELT) – General Data

1. General

The Emergency Locator Transmitter (ELT) is used to locate an aircraft in the event of an accident.

2. Description

Refer to Figure 1 for details.

A. Artex C406–N Series ELT

An Emergency Locator Transmitter (ELT) is a self contained unit operated by an integral sealed battery pack. The ELT is installed in the aircraft on the lower avionics shelf between station 406.00 and 421.00. The ELT is labelled with an arrow marked DIRECTION OF FLIGHT, to ensure correct installation direction for operation of G-switch in the event of impact.

In the event of impact, the G-switch on sensing deceleration of 4.5 fps ± 0.5 fps with a duration of 11 ± 5 milliseconds will operate within the transmitter and activate its signal. The ELT is connected to an antenna installed on the top fuselage just forward of the fin. The ELT will transmit continually until the G-switch is deactivated or the battery is exhausted. The ELT transmits the standard swept tone on 121.5 and 243.0 MHz. Every 50 seconds for 520 milliseconds (long message protocol) the 406.025 MHz transmitter turns on. During that time an encoded digital message is sent to the COPAS-SARSAT satellite.

The navigation portion of the ELT continually receives GPS derived aircraft position information from the APEX system through the modular avionics unit. If the ELT is activated, the ELT will transmit the last recorded latitude and longitude of the aircraft in addition to data identifying the aircraft.

The information contained in the digital message includes: Serial Number of the Transmitter or Aircraft ID, Country Code, I.D. Code and Position Coordinates (Country specific ELT registration requirements may vary).

When mounted in the aircraft and connected to the electrical connector, the transmitter is armed whenever the ELT switch is selected to OFF, and the cockpit remote switch is selected to ARM.

Note Should the transmitter be activated inadvertently, set the ELT or cockpit switch to the ON position, then back to the OFF/ARM position. The ELT will be deactivated and the G-switch reset. If either switch is in the ON position, the other switch will not reset the ELT.

An audio buzzer is installed adjacent to the ELT. The buzzer sounds whenever the ELT is activated.

The buzzer is powered by the ELT unit and therefore is not dependent upon the aircraft battery for operation. It is not designed to operate continuously but sounds at predetermined intervals, and runs for shorter periods towards the end of battery life.

28 Volt DC power is provided from the left bus through a 1 amp ELT NAV (K5) circuit breaker on the avionics circuit breaker panel refer to 24-00-00, Electrical Power – General Data.

Note If the ELT Assembly is moved to a different aircraft, the ELT assembly must be given a new registration number. (Refer to manufacturer's component maintenance manual for more details).

B. Cockpit Control Panel

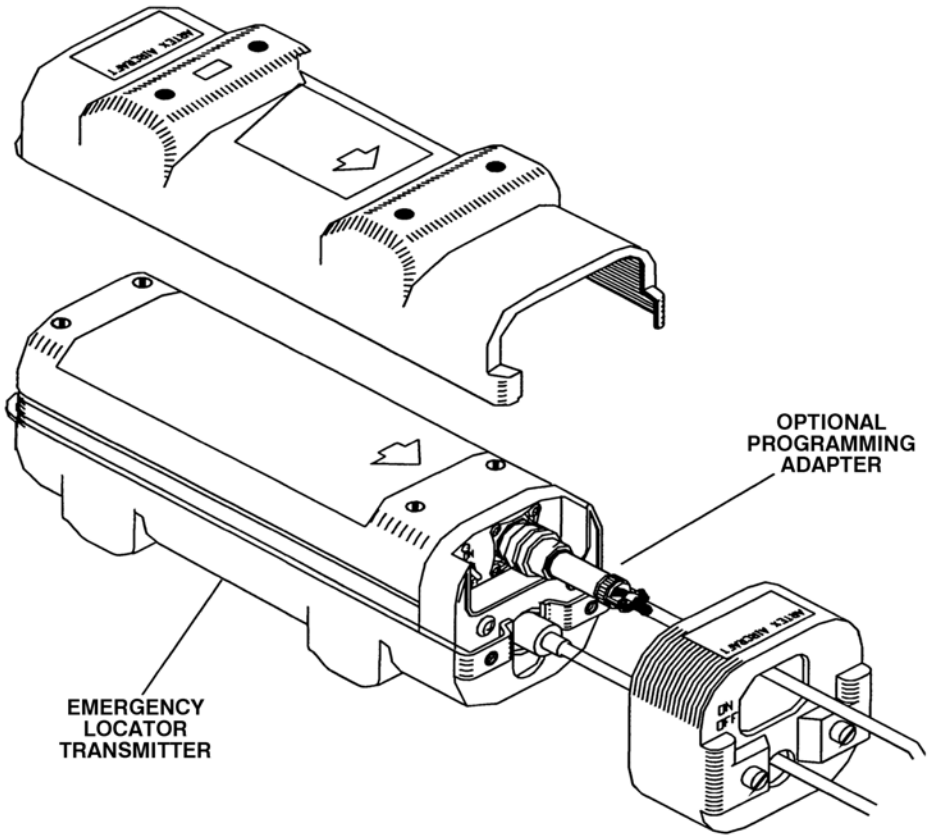
An ELT remote control panel is installed on the outboard side of the right hand sub-panel. This unit has a guarded switch with the positions ON and ARM. If the ELT is accidentally activated, either by tripping the G-switch or with the ELT switches, the indicator light will flash and ELT Active caution CAS message will be displayed on the APEX screen. The indicator light is also used during self test to verify proper operation of circuitry.

28 Volt DC power is provided from the hot bus through a 1 amp ELT SWITCH (U1) circuit breaker on the rear circuit breaker panel refer to 24-00-00, Electrical Power – General Data.

C. Programming Adapter

The Programming Adapter (PA) is an optional accessory for the C406–N series of ELT's. The PA is contained within the ELT's round connector backshell. The PA is approximately 2 inches long (50mm) and has a silver label affixed which records the programming information, refer to Figure 1.

For aircraft with the ELT programming adapter installed, if the ELT Assembly is replaced, the PA will transfer the information held within it to the replacement ELT assembly. (Refer to manufacturer's component maintenance manual for more details).



Emergency Locator Transmitter Assembly
Figure 1

3. Adjustment/Test

A. Transmitter Test

WARNING **DO NOT ALLOW TEST TO EXCEED 5 SECONDS IN DURATION. ACTIVATION OF THE ELT SIGNAL, OUTSIDE OF THE SPECIFIED TIME WINDOW AND LONGER THAN THE 5 SECOND DURATION IS CONSIDERED BY SATELLITE SYSTEMS TO BE A VALID TRANSMISSION SIGNAL.**

Note Always do a transmitter test within the first 5 minutes of the hour (UTC). Ensure you have notified the local control tower, local air traffic of your intentions to do a transmitter test.

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Ensure that ELT NAV (K5) circuit breaker on the avionics circuit breaker panel and ELT SWITCH (U1) circuit breaker on the rear circuit breaker panel are closed.

4) Ensure the ELT switch on the ELT assembly is OFF.

5) Ensure the remote ELT switch, located on the co-pilot's sub panel, is set to ARM.

   Note Ensure the aircraft's navigational system is active at least 30 seconds prior to doing the ELT transmitter test.

6) Tune the one VHF transceiver to 121.5 MHz.

7) Select remote ELT switch to ON.

8) Confirm that the ELT buzzer adjacent to the ELT assembly sounds.

9) Wait approximately 1 second, or for 3 sweeps of the VHF receiver, then select the ELT remote switch to the ARM position, paying special attention to the LED when ARM is selected.

   Note A 406.028 MHz test signal will be transmitted during the reset cycle. It will be coded as a test signal and will not contain position data if viewed with a test set.

10) If the ELT is working properly, the LED will remain ON for approximately 1 second, then extinguish. If a problem is detected, it will be annunciated using a code following the initial pulse, refer to the troubleshooting section, Para 5. for more information.

11) Continue monitoring the VHF transceiver for a few seconds to verify ELT does not continue to transmit.

B. Electrical Bonding Test – ELT Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to empennage through rear baggage compartment.

3) Locate ELT Antenna aft of station 421.00 on the ceiling of the aircraft.

4) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the ELT Antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

5) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

6) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

4. Removal/Installation

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND LIGHTNING. ENSURE INTEGRITY TO THE HIRF/LIGHTNING PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:

SHIELD GROUNDING – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

CAUTION **DO NOT USE CONTACT CLEANER ON ELT COMPONENTS. CHEMICAL CLEANERS CAN BE HIGHLY DESTRUCTIVE TO THE MOUNTING HARDWARE AND ELT HOUSING, AND MAY CAUSE CRACKING, FRACTURING AND BREAKAGE.**

A. ELT

1) Removal of ELT

a) Ensure the aircraft electrical system is OFF.

b) Open the ELT NAV (K5) circuit breaker and fit safety clip.

c) Gain access to the ELT, which is accomplished through the aft baggage compartment into the rear avionics compartment.

d) Loosen the (2) thumb screws which secure the mounting cap to end of ELT assembly.

e) Pull mounting cap away from unit.

f) Lift the top protective cover assembly off the unit.

g) Disconnect antenna coax connector and electrical connector.

h) Lift the connector end of the ELT up to disengage the locking ears at the aft end of the mounting tray. Lift the ELT out of the bottom protective cover.

Note Stow connectors as required.

2) Installation of ELT

CAUTION **IF PROGRAMMING ELT ASSY IS INSTALLED ON AIRCRAFT WHICH HAS A PROGRAMMING ADAPTER INSTALLED, THE DATA HELD IN THE PROGRAMMING ADAPTER WILL OVERWRITE ELT ASSY DATA.**

a) Ensure the aircraft electrical system is OFF.

b) Ensure the ELT NAV (K5) circuit breaker is opened and fitted with safety clip.

c) Reconnect antenna coax connector and electrical connector.

d) Fit the top protective cover assembly onto the unit.

e) Install mounting cap onto the front of unit.

f) Tighten the (2) thumb screws which secure the mounting cap to end of ELT assembly. Do not tighten over 18 in–lbs.

g) Close access panels as required.

h) Remove safety clip and close the ELT NAV (K5) circuit breaker.

i) Do a ELT transmitter test, refer to Para 3.A.

B. ELT Battery Pack

CAUTION **THE BATTERY PACK CONTAINS STATIC SENSITIVE PARTS, TAKE ESD PRECAUTIONS BEFORE HANDLING.**

1) Removal of ELT Battery

a) Remove the ELT, refer to Para 4.A.1).

b) Remove the (4) screws from the bottom of the ELT.

c) Lay ELT on side, carefully lift battery pack away from ELT unit, set it alongside the ELT.

d) Carefully disconnect the 8-pin connector on the small circuit board in the battery pack. Do not short connector pins.

e) Disconnect 2-pin harness from the ELT body.

f) Inspect battery pack for signs of damage, wear and corrosion including the battery cells, the pins, around the circuit board area and outside surface of ELT unit.

g) Inspect the integrity of the wires and connections. Ensure none are damaged or broken.

2) Installation of ELT Battery

a) Connect the 8-pin connector to the header on the circuit board in the battery pack.

b) Connect the 2-pin power harness to the ELT. This will cause the ELT to activate.

c) Reset the ELT by toggling the switch on the ELT from ON to OFF.

d) Lay wires away from the standoffs to avoid pinching wires between standoffs and battery pack.

e) Install (4) screws on the bottom of the ELT.

f) Install the ELT, refer to Para 4.A.2).

C. ELT Mounting Tray

1) Removal of ELT Mounting Tray

a) Remove the ELT, refer to Para 4.A.1).

b) Remove (4) screws and washers holding the mounting tray to the equipment shelf.

c) Remove the mounting tray from the aircraft.

2) Installation of ELT Mounting Tray

a) Place the ELT mounting tray on the equipment shelf.

b) Install (4) screws and washers.

c) Install the ELT, refer to Para 4.A.2).

D. ELT Antenna

1) Removal of ELT Antenna

a) Ensure the aircraft electrical system is OFF.

b) Open the ELT NAV (K5) circuit breaker and fit safety clip.

c) Gain access to the ELT, which is accomplished through the aft baggage compartment into the rear avionics compartment.

d) Place the ELT switch in the OFF position and remove ELT electrical connector to ensure the ELT is not transmitting.

e) Carefully cut the sealant from around the base of the antenna.

f) Remove the (4) screws from the base of the antenna.

g) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

h) Disconnect the coaxial connector.

i) Remove antenna from aircraft.

2) Installation of ELT Antenna

a) Ensure the aircraft electrical system is OFF.

b) Ensure the ELT NAV (K5) circuit breaker is opened and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) A layer of anti-corrosion bonding grease should be applied between the aircraft and the base of the antenna.

f) Secure the antenna assembly on fuselage with (4) mounting screws.

g) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

h) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

i) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching affected area.

j) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

k) Check the antenna electrical bonding as per Para 3.B.

l) Connect the electrical connector and ensure ELT is not transmitting.

m) Remove the safety clip and replace the ELT NAV (K5) circuit breaker in the avionics circuit breaker panel.

n) Do a ELT transmitter test, refer to Para 3.A.

5. Troubleshooting

A. Flashing LED Code

If a problem is detected within the operation of the ELT when the unit is reset, the LED on the remote switch, located on the co-pilot's sub panel, will display a code sequence of flashing lights. Please note that if there are more than one error type detected the coded LED sequence will be displayed in order of importance with approximately a 1 second pause between each coded sequence.

Coded LED Error sequence in order of importance:

1 flash – Indication of G-switch loop open failure.

3 flashes – Indication of a transmission failure of 406.028 Hz signal. May be caused by bad or unconnected coax, an antenna problem or low power output.

5 flashes – Indication of invalid data or no navigational data being received by the ELT. May be caused by improper wiring, improper programming, or invalid navigation data.

7 flashes – Indication of a battery problem. May be caused by battery usage time over an hour.

There is an error condition where the LED on the remote switch will cycle of 1 short and 1 long light signal. This occurs when power is applied to the ELT assembly, and the jumper between pins 3 and 4 is not installed.

B. ELT

1) Ensure ELT NAV (K5) and ELT SWITCH (U1) circuit breakers are closed.

2) Visually inspect ELT and ELT switch connectors for damaged wiring or contacts. Ensure electrical connector on ELT and ELT switch are fully seated and locked.

3) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

4) If repair or service is required, send to appropriately rated avionics repair facility.

6. Servicing

A. ELT Battery Pack

1) Change the battery pack on any of the following conditions:

a) After use in an emergency.

b) Inadvertent activation of unknown duration.

c) When the total of all known transmissions exceeds one hour.

d) On or before the battery replacement date entered on the case of the transmitter (beacon).

e) There is any evidence of corrosion or leakage of any cell or on the small interface board and connector.

Note Additional maintenance and inspection requirements for the installation of the ELT may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

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